

CHEM 3A – Lab Practical – VERSION ASection #: 45598Name: Nia JuvalaStudent ID #: 0754151

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

Density =

0.9638 g $V = 9.9619$

Dry F = 9.6015 g

 $D = m/V$ $W = 19.5634 \text{ g}$

Station #2

85.8%

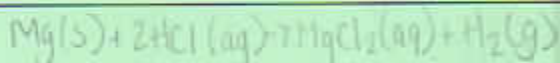
Perc Yield = $Ac / Th \times 100\%$

Known - Alum = 8.819 g

Actual = 7.4415 g

Theo = 8.819 g

Station #3

 $H = 1.008$

Find Mass if 11 m grams if

used: 0.0475 g
piece Mg

Station #4

21.8°C

21.8°C water temp

Station #5

 $V = l \times w \times h$ 5.772 cm³5.8 cm³

length - 0.7

width - 1.2

height - 1.3

cm³

Station #6

Observations: The solution turns to milky-like consistency

Reaction Type: Neutralization

Balanced Equation: $\text{CaCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3 + 2\text{NaCl}$

Station #7

Final = 19.80 mL

$$D = m/v \\ \downarrow \\ = 0.09513$$

V = D/M
(calc the volume)

$$V = 10.00 \text{ mL} \\ \text{Initial} = 1.27 \text{ mL} \\ \text{conc} = 0.9513 \text{ M}$$

Station #8

- Graduated cylinder
- Bunsen burner

Station #9

Unknown - Final molarity
Known: dilute 25.00 mL;
2.5789 M NaCl
Final V = 100.00 mL
10.9156

Station #10

0.1964 g

Sub
↓

$$1.0x = 0.1964$$

Cruc + lid = 21.8503
Cruc + lid + Mg = 21.9751
Cruc + lid + Mg + unknown = 22.1715 1st
22.1722 2nd

Extra Space

CHEM 3A – Lab Practical - VERSION C

Section #: _____

Name: Haniza KhanStudent ID #: 1043694

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$$= \frac{0.0856 \text{ mol}}{0.05 \text{ L}}$$

$$M = 1.712 \text{ mol/L}$$

$$\begin{aligned} \text{50 mL} \\ = 0.05 \text{ L} \end{aligned}$$

$$= 0.0856 \text{ mol}$$

Station #2

$$Al = 0.3301 \text{ g}$$

$$0.3301 \times \frac{474.46}{26.98} = 5.81 \text{ g alum}$$

Station #3

$$T = 18.5^\circ\text{C}$$

$$P_{\text{total}} = 763.00 \text{ mmHg}$$

$$P_{\text{H}_2} = P_{\text{H}_2\text{O}}$$

Station #4

$$q = 35.03 \text{ g} \times 4.184 \text{ J/g}^\circ\text{C} \times 21.1^\circ\text{C}$$

$$q = 3093 \text{ J}$$

Station #5

water +

rectangular
cube

$$= 37.9 \text{ mL}$$

$$V = 7.9 \text{ mL}$$

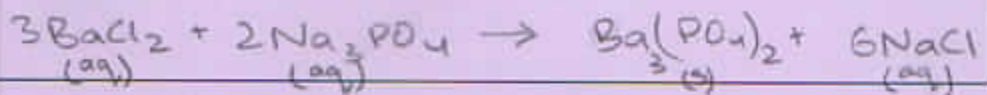
Station #6

$$\text{Na}_3\text{PO}_4, \text{BaCl}_2$$

Observations: solution became milky & grainy, like expired milk.

Reaction Type: Double disp.

Balanced Equation:



Station #7

5.2 mL

17.5 mL

Station #8

eye piece (glass piece), pipet

Station #9

3.9cm

Station #10

Mass % of Mg = 60%

$$Mg = 0.1495g$$
$$Mg^{+0}y = 0.2479g$$
$$O_2 \text{ in } MgO_y = 0.0984g$$

Extra Space

CHEM 3A – Lab Practical - VERSION B

Section #: _____

Name: Rafaelo Marmilla MartinezStudent ID #: 0977963

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

$$\frac{51.9349 \text{ g}}{100.11417} = 51.87\%$$

50 mL

Station #2

Vial: 17.2665

Second: 27.1941

Together: 44.4607g

$$\begin{array}{r} 27.1941 \\ - 17.2665 \\ \hline 9.92762 \end{array}$$

Theoretical yield

Station #3

Measure: 41.60 mL

Station #4

The temperature went down during the reaction

It was exothermic

Station #5

longest dimension in cm = 3.60 cm

Convert to mm

36.0 mm

Station #6

Observations: liquid is clear, gotten slightly warmer

Reaction Type: No reaction

Balanced Equation: $\text{HCl} + \text{Na}_2\text{CO}_3 = 2\text{NaHCO}_3 + \text{H}_2\text{O}$

Station #7

1.585

V: 10.00 I: 12.7 F: 17.12 mL MSICS

Station #8

Volumetric Flask:

Test tube

Station #9

Temperature: 21.5°C

Volume of water: 20.60 mL

Station #10

$$\frac{0.1003\text{g}}{0.2577\text{g}} = 0.3969 \times 100\% = 39.69\%$$

mass of MS: 0.1529g

Extra Space

CHEM 3A – Lab Practical – VERSION A

Section #: _____

Name: Miguel NegreteStudent ID #: 0977589

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

Empty flask 9.5895 g L with 10mL solution including flask

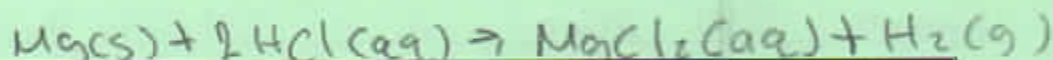
$$\frac{20.0994 \text{ g}}{10.0} = 2.00994 \text{ Density}$$

Station #2

$$\frac{7.4415}{8.879} \times 100 = 83.810\% \text{ Percent Yield}$$

That is a reasonable amount its within 80-100% mark.

Station #3



Calculate the mass of hydrogen gas (in g) that could be made if a student used a 0.0475g piece of Magnesium.

$$0.0274 \text{ g made} \quad \begin{array}{l} \text{Mg} - 0.0475 \\ \text{Cl} - 35.453 \times 2 \\ \text{H} - 2 \end{array}$$

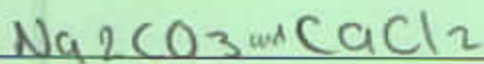
Station #4

21.4°F was the reported amount

Station #5

$$L = 3.5 \times 1.0 \text{ w} \times 3.5 \text{ H} = 12.25 \text{ cm}^3$$

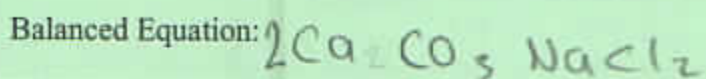
Station #6



Observations: The mixtures combined became "milky" and leaving residue on the tube

Reaction Type: Precipitation

Balanced Equation:



Station #7

0.9573 M

Volume of Column
10. mL

19.8 mL was final reading

$19.8 - 11.27 = 8.53 \text{ mL}$ was the amount used

Station #8

A Bunsen burner and a 25 mL beaker

Station #9

Final Molarity of a solution made by diluting 25.00 mL of 2.5789 M

$$\begin{array}{ccccc} \text{Diluting} \nearrow & 25.00 \text{ mL} & 2.5789 \text{ M} & = & 100.00 \text{ mL} & \frac{(25)(2.5789)}{100} \\ & 25.00 \text{ mL} & \nearrow \text{mole} & & \nearrow \text{Final} & = 0.644725 \text{ M} \end{array}$$

Station #10

Between 0.1964 g - 0.1971 g of oxygen

Extra Space

CHEM 3A – Lab Practical - VERSION B

Section #: _____

Name: Amiel HernandezStudent ID #: 1038679

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

$$\frac{M}{V} \% = \frac{\text{mass Solute (g)}}{\text{mL Solution}} \times 100$$
$$= \frac{5.0277}{48.1803} \times 100 = \boxed{10.44 \%}$$

Station #2

Empty: 17.2662

27.1944

Full: 27.1944

$$\begin{array}{r} 27.1944 \\ - 17.2662 \\ \hline 9.9282 \end{array}$$

Mass Alum: 9.9282 g
theoretical yield

Station #3

40.4 mL

Station #4

Temp went down
Endothermic

Station #5

3 1/2 cm →

Station #6

Observations: No Change in Color

Reaction Type: NO Reaction

Balanced Equation: $\text{HCl}_{(aq)} + \text{Na}_2\text{CO}_{3(s)} \rightarrow \text{HCl} + \text{Na}_2\text{CO}_3$

Station #7

Volume of down: 10 mL

Initial: 1.27 mL

Final: 17.12

Molarity = (mol Solute) ÷ (L Solution)

Station #8

Test tube
flask

Station #9

Volume of water: 28.0 mL

Temp: 21°C

Station #10

Mass Mg used: 0.1524

Mass of Mg_xO_y : 0.2527

Mass of Oxygen in Mg_xO_y : 0.1003

What is Percent of oxygen:

39.69%

Extra Space

CHEM 3A – Lab Practical – VERSION A

Section #: _____

Name: Ashley ArellanoStudent ID #: 1016534

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

$$\text{Density} = \frac{\text{mass}}{\text{volume}} \quad \frac{25.2696\text{g}}{10.00\text{ mL}} = 2.52696$$

Station #2

Calculate Percent Yield Alum 8.879g 7.4415g
reasonable value

$$\text{PY}(\%) = (\text{ay}) / (\text{ty}) \times 100\% = 8.879\text{g} / 7.4415\text{g} \times 100\%$$

$$\text{Percent yield} = 119.32$$

Station #3

$$\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2(\text{g})$$

mass of hydro (g)
0.0475g mag
Avg mass = 1.008

$$\frac{0.0475\text{g}}{1} \times \frac{1.008\text{g}}{1\text{g}} = \frac{0.0475\text{g}}{1.008} = 0.047$$

Station #4

25 mL plastic cup, styrofoam cup take temp
= 20.2°C

Station #5

L = 3.7 cm
h = 1.3 cm
w = 1.3 cm

$$V = L \times w \times h$$

$$3.7 \times 1.3 \times 1.3 =$$

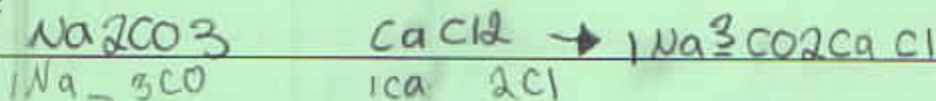
$$V = 6.253\text{ cm}^3$$

Station #6

Observations: cloudy, gray/white, sizzling

Reaction Type: combustion

Balanced Equation:



Station #7

NaOH 0.9573m

v soda = 10.00 mL

Initial buret = 1.27 mL

Final buret = 8.73 mL

$$\begin{array}{r} 10.00 \text{ mL} \\ - 1.27 \text{ mL} \\ \hline 8.73 \end{array}$$

10.00 mL - 1.27 mL

Station #8

Beaker - to measure any amount of fluid in mL

? = connect the gas tube, flame will come out

Station #9

Final molarity

25.00 mL of a 2.5789 M sodium

$$\frac{0.25}{2.5789} = 0.0969$$

Final sol = 100. mL

Station #10

m ec + Lid = 21.8563 g

mass = 15.44 g

m cb + Lid + mg = 21.9751 g

m cb + Lid + mg + CP = 22.1715 g
22.1722 g

= 0.0007

0.0001

mass oxygen reacted?

$$\begin{array}{r} 22.1722 \\ - 22.1715 \\ \hline 0.0007 \end{array}$$

Extra Space

CHEM 3A - Lab Practical - VERSION CSection #: ThursdayName: Belen HernandezStudent ID #: 0982244

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$$\frac{5.0234 \text{ g NaCl}}{1 \text{ mol}} \cdot \frac{1 \text{ mol}}{58.44277 \text{ g/mol}} = 0.160463$$

$$58.44277 = (5.0234) / (50.0 \text{ mL}) = \boxed{0.1 \text{ M}}$$

Station #2

$$\begin{aligned} \text{Aluminum} &= 0.330 \text{ g washed} \\ \text{Aluminum} &= 26.98 \text{ g/mol} \\ \text{Alum}(\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}) &= 474.46 \text{ g/mol} \end{aligned}$$

$$2\text{Al} + 2\text{KOH} + 9\text{H}_2\text{SO}_4 + 22\text{H}_2\text{O} \rightarrow 2\text{KAl}(\text{SO}_4)_2 + 12\text{H}_2\text{O} + 3\text{H}_2$$

$$551.044 + 493.46 = \boxed{57.98}$$

Station #3

$$\begin{aligned} \text{Temp of water} &= 18.09^\circ \\ \text{Pressure} &= 763.1 \\ \text{Pressure of hydrogen gas} &= 42.0 \end{aligned}$$

$$P_{\text{total}} = P_{\text{H}_2} + P_{\text{H}_2\text{O}}$$

$$763.1 = 18.09^\circ + 42.0 = 62.09$$

$$\boxed{62.09}$$

Station #4

$$Q = m \times c \times \Delta t$$

$$Q = 35.0392 \text{ g} \times 4.184 \text{ J/g}^\circ\text{C} \times 21.1^\circ = 3093.34 \text{ J}$$

$$\boxed{Q = 3093.34} \quad 3093.$$

Station #5

$$20 \text{ mL}$$

$$22.0 \text{ mL water} + \text{thimerosal}$$

Station #6

Observations: Cloudy, thick, in chunks

Reaction Type: Combustion

Balanced Equation: $\text{Na}_3\text{PO}_4 \rightarrow \text{BaCl}_2 : \text{Na}_3\text{PO}_4 \rightarrow 3\text{BaCl}_2$

Station #7

20X0

27.07

19.08 ml

Station #8

1. Specimen glass: → holds the chemicals to see the reactions
2. Pipet Wash: holds the pipet from top so that we can use it to suck out any water needed to add to any other container

Station #9

3.08 cm

Station #10

mass mg used: 0.1495g

mg NaOy Combustion: 0.2479g

O₂ in mg NaOy: 0.0984g

$$\frac{0.1495g}{1 \text{ mol}} \cdot \frac{0.2479g}{0.0984g} = 0.3766g$$

0.3771g

Extra Space

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$M = (mol\ solute) / L\ (solution)$
 • 50 mL of sol A.
 $\downarrow$
 • $5.0234\ g\ of\ NaCl \times \frac{1\ mol\ 6.08596}{58.44\ g\ of\ NaCl} = mol$
 $\cdot 50.00\ mL \times \frac{1\ L}{1000\ mL} = 0.05\ L$
 $= \frac{0.08596\ mol}{0.05\ L} = 1.72\ \frac{mol}{L}$

Station #2

• Mass of aluminum foil = $0.3305\ g$
 $\rightarrow$ maximum amount of Al in grams = $0.3305\ g$
 • $0.3305\ g\ Al \times \frac{1\ mol}{26.982\ g} = 0.01225\ mol$
 $\cdot (26.982\ g / 0.3305\ g) = 81.64\ g\ of\ Al$

Station #3

• Temp of $H_2O = 18.50$
 • $(P_{total}) = P_{H_2} + P_{H_2O}$
 $(339\ k) + 18.50 = 358\ K \rightarrow 0.2\ in\ mm\ Hg$
 $= 660 + 18.50 = 339\ k$
 or $(66 + 273) = 339\ k$

Station #4

• $q = m \times C \times \Delta T$
 • $q = (35.0392\ g) \times (4.184\ J/g^\circ C) \times 21.1^\circ C = 309\ J$
 $q = 309\ J$

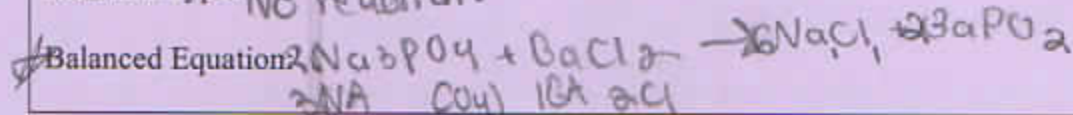
Station #5

• Vol of object C = $4.50\ cm$
 $\rightarrow 4.50\ cm \times \frac{1\ mm}{0.1\ cm} = 45.0\ mm$

Station #6

Observations: It's clear-like that's slightly grayish + white-like color. There's no bubbles or reaction!

Reaction Type: No reaction!

**Station #7**

• 20.0 for the final buret reading!

Station #8

• Watch-glass + volumetric pipet!

Station #9

• 3.90 cm!

Station #10

• mass percentage = $\frac{\text{g element}}{\text{g compound}} \times 100\%$

$\frac{0.1495 \text{ g of Mg}}{0.2479 \text{ g of Mg}_2\text{O}_4} \times 100 = 60.31\%$

Extra Space

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

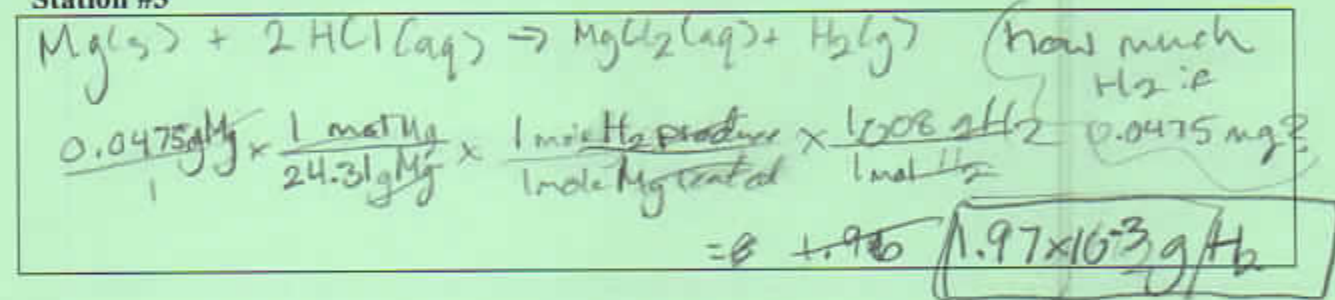
$$\begin{array}{r}
 \text{Flask} - 13.750\text{g} \\
 \text{Flask + water} \rightarrow 24.2495\text{g} \\
 10\text{mL} \\
 \hline
 24.2495 \\
 - 13.7503 \\
 \hline
 10.4992 \\
 \hline
 10.4992 \\
 10.0\text{mL} \\
 \hline
 = 1.0499 \text{ (1.05 mL)}
 \end{array}$$

Station #2

$$\begin{array}{l}
 \text{theoretical yield} = 8.879\text{g} \\
 \text{actual yield} = 7.4415\text{g} \\
 \frac{7.4415\text{g}}{8.879\text{g}} = 83.81\% \text{ percent yield}
 \end{array}$$

This is a reasonable yield. They potentially lost some material during transferring.

Station #3



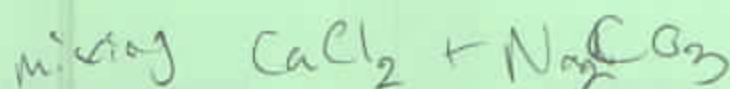
Station #4

$$21.9^\circ\text{C}$$

Station #5 answer in cm^3

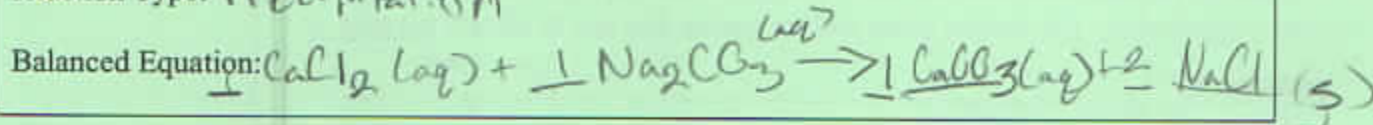
$$\begin{array}{l}
 L = 3.8\text{cm} \\
 W = 1.2\text{cm} \\
 h = 1.3\text{cm} \\
 V = 3.8\text{cm} \times 1.2\text{cm} \times 1.3\text{cm} = 5.928 \\
 \hline
 5.9\text{cm}^3
 \end{array}$$

Station #6



Observations: immediately cloudy white precipitate on walls

Reaction Type: Precipitation



Station #7

Final buret reading = 21.5 mL

Initial buret reading = 1.27 mL

$$\begin{array}{r} 21.5 \text{ mL} \\ - 1.27 \text{ mL} \\ \hline \end{array}$$

20.23 mL NaOH used

Station #8

Bunsen Burner
to heat stuff
up

Graduated Cylinder -
to precisely measure aqueous
solutions

Station #9

final molarity diluting 25.00 mL of 2.5789 M NaCl to 100.0 mL

$$25.00 \text{ mL NaCl} \rightarrow 100.0 \text{ mL}$$

$$\frac{2.5789 \text{ M NaCl}}{100 \text{ mL}} = 0.025789 \text{ M}$$

$$= 0.025789 \text{ M}$$

Station #10

Mass of O reacted

$$\text{Empty crucible + lid} \rightarrow 21.8563 \text{ g}$$

$$+ \text{Mg} \rightarrow 21.9751 \text{ g}$$

$$+ \text{MgO} \rightarrow 22.1715 \text{ g}$$

$$22.1722 \text{ g}$$

$$\begin{array}{r} \text{mg} = 21.9751 \text{ g} - 21.8563 \text{ g} \\ = 0.1188 \text{ g Mg reacted} \end{array}$$

$$\begin{array}{r} 0.3859 \\ - 0.1188 \\ \hline \end{array}$$

$$\begin{array}{r} 22.1722 \\ - 21.8563 \\ \hline 0.3159 \end{array}$$

0.1971 g O reacted

Extra Space

CHEM 3A – Lab Practical - VERSION C

Section #: _____

Name: Tayaniyah BryantStudent ID #: 10165569

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$$m = \frac{(5.0234\text{g})(58.44277\text{ g/mol})}{50.00\text{ mL}} = 5.8716$$

$$= 5.872$$

Station #2

$$0.3303\text{g} = c$$

$$\frac{(26.98\text{ g/mol})(0.3303\text{g})}{474.46\text{ g/mol}} = 0.0188\text{g}$$

Station #3

$$19^\circ \text{ water } 152, 40.9$$

$$19^\circ = 152 + 40.9$$

$$P = 10.7$$

$$P_{\text{total}} = P_{\text{H}_2} + P_{\text{H}_2\text{O}}$$

$$P_{19^\circ} = 152\text{ H}_2 + 40.9$$

$$P_{19^\circ} = 192.9$$

Station #4

$$q = 35.0392\text{g} \times 4.184\text{ J/g}^\circ\text{C} \times 21.1^\circ\text{C} + 30.25$$

$$= 5763.887$$

$$= 576\text{g}$$

Station #5

$$37.0\text{ mL}$$

Station #6

Observations: reaction turned milky white, not fully dissolved

Reaction Type: acid / Base neutralization

Balanced Equation: $\text{Na}_3\text{PO}_4 + \text{BaCl}_2 \rightarrow \text{NaBa} + 3\text{PO}_4\text{Cl}_2$

Station #7

20.00 mL

Station #8

A) graduated cylinder
B) Volumetric pipet
watch glass

Station #9

3.9 cm

Station #10

$$\text{mass \%} = \frac{0.1495\text{g}}{0.247\text{g}} \times 100 = 61\%$$

Extra Space

CHEM 3A - Lab Practical - VERSION BSection #: 1Name: Allison Garcia-HernandezStudent ID #: 1063699

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

$$\frac{46.9667}{51.9311\text{g}} \times 100 = 90.32\%$$

Station #2

mass of the Alum

$$26.98\text{g/mol}$$

Station #3

Volume of gas
collected = 40 L

Station #4

The temp went down.

The reaction was endothermic

Station #5

Product B

$$1\text{ cm}^3 = 1,000\text{ mL}$$

Station #6

Observations: bubbles formed

Reaction Type: Acid/base neutralization

Balanced Equation: $\text{HCl} + \text{Na}_2\text{CO}_3$

Station #7

$$\text{molarity (M)} = \frac{17.12}{10.00 \text{ mL}} = 1.71$$

Station #8

1) flask / buret?

2) test tube

Station #9

1) 28 mL

2) 22°C

Station #10

$$0.2527 - 0.1003 = 0.1524 \text{ Mg} \times 0.9 = 15.1\%$$

Extra Space

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

50.00 mL / 5.0234 g NaCl

$$\text{Molarity (M)} = (\text{mol solute}) / (\text{L solution})$$

$$m_m = 58.4427$$

$$5.0234 \text{ g NaCl} / 50.00 \text{ mL}$$

$$\text{Molarity} = 0.100468$$

$$= 0.100$$

Station #2

0.3303 g → mass of Alum foil calculated

$$\text{Alum foil (Al)} = 26.98 \text{ g/mol}$$

$$\text{Alum (Al(SO}_4)_3 \cdot 12\text{H}_2\text{O)} = 474.46 \text{ g/mol}$$

$$\frac{474.46 \text{ g/mol}}{26.98 \text{ g/mol}} = 17.5856$$

$$\text{or } 17.6$$

Station #3

Temp of water = 18.4°

$$P_{\text{total}} = 762.8$$

$$P_{\text{total}} = P_{\text{H}_2} + P_{\text{H}_2\text{O}}$$

$$15.871$$

in book

Station #4

$$q = m \times C \times \Delta T$$

$$35.0392 \text{ g} \times 4.184 \text{ J/g}^\circ\text{C} \times 21.1^\circ\text{C} = 3093.34467$$

↓

$$q = 3093.3450$$

Station #5

25.60 mL is the measurement
for the perm.

Station #6

Observations: When mixing these 2 solutions they became a cloudy white color and powdery texture

Reaction Type: Acid/Base Neutralization

Balanced Equation: $\text{BaCl}_2 + \text{Na}_3\text{PO}_4$

Station #7

19.80 mL is the final buret reading.

Station #8

The lab equipment is a pipet. This is used to help transfer another liquid into another lab equipment, such as into a flask when conducting an experiment.

Station #9

The length of the line is 3.70 cm.

Station #10

Mass of Hg used - 0.1495 g

mass of MgSO₄ - 0.2479 g

mass of MgSO₄ - 0.6984

$\frac{m}{v}$ /

mass percent of magnesium

$$\left(\frac{0.0984 \text{ g}}{0.2479 \text{ g}} \right) \times 100\% \quad \text{Mass percent}$$

$$0.3969 \times 100\% = 39.69\%$$

Extra Space

CHEM 3A - Lab Practical - VERSION CName: Socorro Castro

Section #: _____

Student ID #: 0759215

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

To make 50.00 mL of solution A a student weighed 5.0234 g of NaCl. calculate the molarity of solution A.

The molar mass of NaCl is 58.44277 g/mol

molarity (M) = (mol solute / L solution) 1.72 M

Station #2

0.3304g alum foil

0.01224g

1436.01694g

Station #3

Temp - 19°C

Pressure - 762.7

196.09

Station #4

 $35.0392\text{g} \times 4.184\text{ J/g}^\circ\text{C} \times 21.1^\circ$ 3091.6 J

Station #5

20 mL

Station #6

Observations: When mixed the products turned milky

Reaction Type: Redox

Balanced Equation:

Station #7

-20.0 mL

Station #8

glass pipette & watch
glass

Station #9

3.9 cm

Station #10

1 mass of Mg used 0.1495 g 60.27%

2 mass of the MgO compound 0.2479 g

mass of Oxygen in MgO compound 0.0984 g

Using the experimental data what is the mass computed percent
of magnesium in the MgO sample

Extra Space

CHEM 3A - Lab Practical - VERSION ASection #: ThursdayName: Jovanna GarciaStudent ID #: 0778750

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

$$\rho = \frac{m}{V}$$

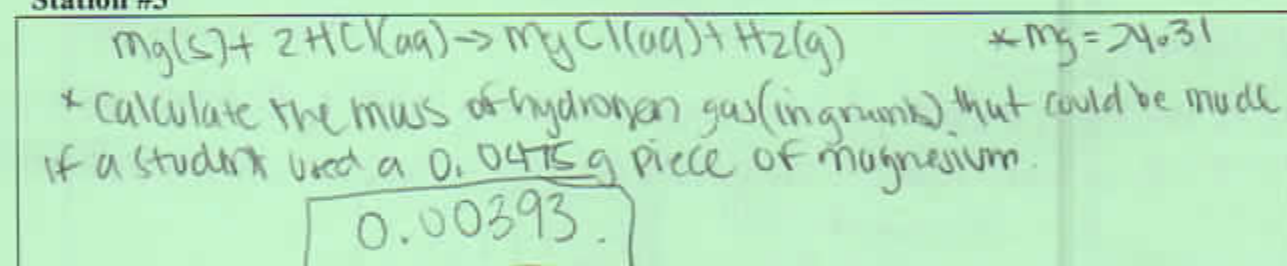
$$\frac{9.5228g}{17.5331g} = \frac{0.5431g}{V}$$

Station #2

Percent yield % = $\frac{(\text{Actual yield})}{(\text{Theoretical yield})} \times 100\%$

$$\frac{8.879g}{10.71375g} \times 100 = 83.81\%$$

Station #3



Station #4

$$22^\circ C$$

Station #5

$$V = L \times W \times h$$

$$3.7 \times 1.3 \times 1.3 = 6.3 \text{ cm}^3$$

Station #6

Observations: Na_2CO_3 was a clear liquid inside the tube when adding CaCl_2 the liquids combined together became a milky white color

Reaction Type:

Acid based Neutralization

Balanced Equation:

Station #7

15.87 mL

$$\begin{array}{r} 0.9513 \\ - 1.27 \\ \hline \end{array}$$

Station #8

Burner = used for gas to be transmitted to produce fire. The burner can be adjusted to create how much fire is needed. 25 ml graduate cylinder. Used for liquid measurements in milliliters.

Station #9

25.00 mL x _____

* Sodium 22.990

what is the final molarity of a solution made by diluting 25.00 ml of 2.5789 M sodium chloride solution to a final volume of 100.00 mL?

$$\frac{25.00 (2.5789)}{100.00}$$

0.6447 M

Station #10

~~1.1971~~

-0.1971

Extra Space

CHEM 3A - Lab Practical - VERSION B

Section #: _____

Name: Destiny BecerraStudent ID #: 1070162

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1 Calculate the mass-vol % of SOL A (NaCl) made in 50.00 mL

$$\left(\frac{M}{V}\right) \% = \frac{\text{mass Solute g}}{\text{mL Sol}} \times 100$$

$\text{SOL A} = 53.208 \text{ g}$
 50.00 mL

$\text{Ans} = 106.416\%$
 $\text{All} = 1.9344$
 $\text{SOL} = 1147\%$

1.064 %

Station #2 Determine the mass of the alum

empty - 17.266/g
 Alum = 26.982 amu
 $\times 1.6605 \times 10^{-24}$

44.804 x 10⁻²⁴

Station #3

Temp 19.4°C
 P. 7.62.8
95.21 amu

Station #4

endothermic, the temperature went down

Station #5

$3.8 \text{ cm} = 38 \text{ mm}$
 $\times 10$
38
 $1 \text{ cm} = 10 \text{ mm}$

Station #6

Observations: When I poured the solutions into the ^{test} ~~container~~ there was no reaction

Reaction Type: No reaction

Balanced Equation: $\text{HCl} + \text{Na}_2\text{CO}_3$

Station #7

0.9573 m

$$\begin{array}{r} \text{soda} \\ \times 10.00 \text{ mL} \\ \hline \text{NaOH} 1.27 \text{ mL} \\ \hline 12.7 \text{ mL} \end{array}$$

Station #8

Testing tube & Flask

Station #9

27.5 mL
Cylinder

22.5 °C
Flask

Station #10

$\text{Hg} = 0.1524 \text{ g}$

$\text{Mg} \times \text{O}_y^{\text{comp}} = 0.257 \text{ g}$

$\text{O}_2 = 0.1003 \text{ g}$

Mass % O_2 in $\text{mg} \times \text{O}_y$

0.0015 %

Extra Space

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

$$\begin{aligned} \text{Flask} &= 10.2894 \text{ g} & D &= \frac{m}{V} & \frac{26.8397 \text{ g}}{16.2894 \text{ g}} & \frac{10.2894 \text{ g}}{10.00 \text{ mL}} \times \frac{1000 \text{ mL}}{1 \text{ L}} \\ 10.00 \text{ mL} &= \text{volume} & & & & \\ \text{Final Flask + Sol} &= 26.8397 \text{ g} & & & & = \boxed{1.05503 \text{ g/mL}} \end{aligned}$$

Station #2

$$7.4415 \text{ g} / 8.879 \text{ g} \times 100 = \boxed{83.81\%}$$

It is reasonable

Station #3

$$\begin{aligned} M_g &= 0.0475 \text{ g} \\ H &= \boxed{0.3950 \text{ g}} \end{aligned}$$

Station #4

$$\boxed{21.80^\circ \text{C}} \text{ temp of water}$$

Station #5

$$V = l \times w \times h$$

$$3.7 \text{ cm (l)} \times 1.2 \text{ cm (w)} \times 1.3 \text{ cm (h)} = \boxed{5.77 \text{ cm}^3}$$

Station #6

Observations: white substance, milky

Reaction Type: Acid / base neutralization

Balanced Equation: $\text{CaCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{NaCl} + \text{CaCO}_3$ **Station #7**19.9 final reading initial ~~1.2~~ 1.27 mL

Final buret = 19.90 mL

Initial buret = 1.27 mL

19.9 mL
1.27

18.63 mL

Station #8Bunsen burnerGraduated cylinder**Station #9**

25.0 mL of 2.5789 M

final vol = 100.0 mL

$$\frac{25.0 \text{ mL}}{2.5789 \text{ M}} \times \frac{1 \text{ M}}{100.0 \text{ mL}} = 0.09694$$

0.09694 M

Station #10

Mass of crucible + lid + Mg = 21.9751 g

Mass of lid + Mg + O₂ compound = 22.1715 g

Mass Oxygen used = 0.1964 g

Extra Space

CHEM 3A – Lab Practical – VERSION ASection #: 43958Name: Jesus San "Roman"Student ID #: 0547169

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

$$\begin{aligned} \text{Dry flask} &= 9.5897 \\ \text{Flask w/ 2 mL} &= 11.5532 \text{ g} \\ &\quad - 9.5897 \\ &\quad \hline &\quad 1.9635 \text{ g} \\ &\quad \hline &\quad 2 \text{ mL} \end{aligned}$$

$$\text{Density} = \frac{m}{V}$$

$$\frac{1.9635 \text{ g}}{2 \text{ mL}} = 9.8175$$

$$\boxed{9.8 \text{ g/mL}}$$

Station #2

$$\begin{aligned} \text{Theoretical Yield} &= 8.879 \text{ g} \\ \text{Actual Yield} &= 7.4415 \text{ g} \end{aligned}$$

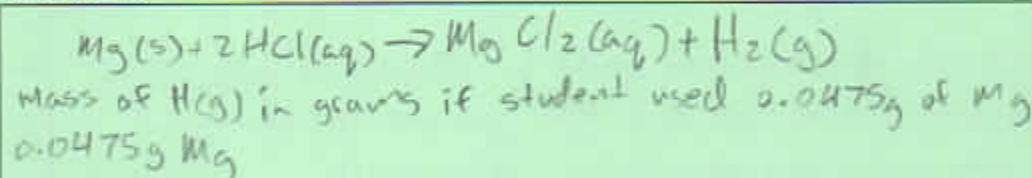
$$\frac{7.4415 \text{ g}}{8.879 \text{ g}} = 0.838101138 \times 100\%$$

$$83.8101138\%$$

$$\boxed{83.8\% \text{ yield}}$$

The value is reasonable because some materials could be lost during the reaction.

Station #3



Station #4

21°C

Station #5

Volume of Object/Version A

$$V = l \times w \times h$$

$$3.7 \times 1.3 \times 1.3 =$$

$$3.7 \times 1.3 \times 1.3 = 6.253 \text{ cm}^3$$

$$\boxed{6.3 \text{ cm}^3}$$

Station #6

Observations: $\text{CaCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow$ Milky white substance formed with fine granules

Reaction Type: Precipitate

Balanced Equation: $\text{CaCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow 2\text{NaCl} + \text{CaCO}_3$

Station #7

NaOH Concentration = 0.9573 M Trial 1
Vol 64 = 10.00 mL
Initial B = 1.27 mL NaOH
Final B = 21.11 mL NaOH
Volume used = 21.11 - 1.27 = 19.84 mL NaOH

Station #8

25 mL graduated cylinder
Bunsen Burner

Station #9

What is the final molarity of a solution made by diluting 25.00 mL of a 2.5789 M sodium chloride solution to a final volume of 100.00 mL?

$$M_1 V_1 = M_2 V_2 \quad (2.5789 \text{ M NaCl})(25.00 \text{ mL}) = (x)(100.00 \text{ mL})$$

$$0.644725$$

$$0.64473 \text{ M NaCl}$$

Station #10

Mass of empty crucible + Lid = 21.8563 g

Mass of crucible + Lid + Mg = 21.9751 g

Mass of crucible + Lid + MgO compound = 22.1715 g 1st weighing

+ 22.1722 g 2nd weighing

$$= 44.3437$$

$$= 22.1715 - 21.9751 = 0.196750$$

Extra Space

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

$$\frac{51.9344\text{g}}{100.1147\text{g}} \times 100 = 51.87489949$$

$$5.0277 \quad \frac{5.0277 \times 100}{53.208} = 9.449142986 = \boxed{9.45\%}$$

Station #2

empty 17.2665g
Full Alum 27.1941g

$$27.1941 - 17.2665\text{g} = 9.9276\text{g}$$

$$= 9.97\text{g Al}$$

actual yield
9.97g Alum

Station #3

19°C degrees
763.1 pressure

Mg 24.31
H 1.008
Cl 35.45

4.70 ml

16.477

$$= \boxed{25.223}$$

Station #4

Temperature went up
exothermic

Station #5

3.5 cm - Cylinder

$$\frac{3.5}{1} \times \frac{10}{1} = \boxed{350 \text{ mm}}$$

mm = 10 = 1cm

Station #6

Observations: Clear, small bubbles formed

Reaction Type: Precipitation

Balanced Equation: $\text{HCl} + \text{Na}_2\text{CO}_3$ **Station #7**

6 Down 10.00 ml

initial 1.2 ml

Final 17.12 ml

0.9573 M

0.5841 Mol Citric acid

$$M = (\text{mol solute}) / (\text{L solution})$$

 $\text{H}_3\text{C}_6\text{H}_5\text{O}_7$ 6 Down molarity of citric acid~~0.5841 M~~**Station #8**Cleymer Flask
test tube

not cure of spelling

Station #9

graduated cylinder

28.50 ml

Flask

21.5°C

Station #1024.31 g
13.999 g.2527 g
.1003 g

= 2.519441675

= 0.025194417

= 0.025% Oxygen

Extra Space

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$$\text{Sol } a \text{ (M)} = \frac{5.0234 \text{ g}}{50.0 \text{ mL}} = \boxed{0.100468 \text{ or } 1.0045 \times 10^{-1}}$$

$$M = \frac{\text{mol solute}}{\text{L solution}}$$

Station #2

Mass of Al: 0.3304 g

$$\frac{0.3304 \text{ g}}{26.98 \text{ g/mol}} = \boxed{0.0122 \text{ g}}$$

Station #3



Temp of water: 18.0°C

Atm. pressure: 762.9 mmHg

$$P_{\text{total}} = 762.9 \text{ mmHg} + 15.477 = \boxed{778.377 \text{ mmHg}}$$

Station #4

$$q = m \times C \times \Delta T = 35.0392 \text{ g} \times 4.184 \text{ J/g}^\circ\text{C} \times 21.1^\circ\text{C} = \boxed{3093.34}$$

Station #5

final volume: 27.0 mL

initial volume: 20.0 mL

Station #6

Observations: milky white color,

Reaction Type: combustion

Balanced Equation: $1 \text{ Na}_3\text{PO}_4 + 1 \text{ BaCl}_2 \rightarrow 1 \text{ BaPO}_4 + 1 \text{ Na}_3\text{Cl}_2$

Station #7

22.00 mL

Station #8

Item #1: watch glass - used to cover boiling water

Item #2: used to transfer liquid with

Station #9

3.9 cm

Station #10

$$\text{mass \%} = \frac{0.2479}{0.1495} = 1.6582 \%$$

Extra Space

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

10.00 mL

need to find density

empty: 9.4294g

filled: 19.8991g

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$d = \frac{9.4294g}{19.8991g} = \text{density} = 0.47386g$$

Station #2

8.879 g.

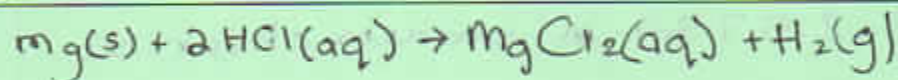
mass = 7.4415g

$$(7.4415g) / (8.879g) \times 100\%$$

Percent Yield = 83.8%

$$0.83810 \times 100\% = 83.8$$

Station #3



0.0475g Mg

:-?

Station #4

temperature = 21.9 °C

or

temperature = 21.90 °C

Station #5

$$V = L \times W \times h$$

$$3.7cm \times 1.2cm \times 1.2cm = V$$

$$V = 5.328cm^3$$

Station #6

Observations: the color changed from clear to milky white

Reaction Type: acid/base neutralization

Balanced Equation: $\text{Na}_2\text{CO}_3 + \text{CaCl}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{CaCl}_2$
 ~~$\text{Na}_2\text{CO}_3 + \text{CaCl}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{CaCl}_2$~~

Station #7

$$10.00^{\text{mL}} + 1.27 \text{ mL} = \text{Volume}$$

$$\text{Volume} = 11.27 \text{ mL}$$

Station #8

bunser burner

~~25 mL beaker~~

25 mL graduated cylinder

sorry for the spelling errors

Station #9

25.00 mL of 2.5789 M sodium chloride

100.00 mL

?? ?

Station #10

$$22.1722 \text{ g} - 22.1715 \text{ g} = 0.0007 \text{ g}$$

$$\text{mass of oxygen} = 0.0007 \text{ g}$$

Extra Space

CHEM 3A - Lab Practical - VERSION C

Section #: _____

Name: Osce JimenezStudent ID #: 0958268

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

50.00 mL solution
 5.0234 g NaCl
~~mol~~ $m = 0.085954 \times (0.0502)$
Mr. 6.642977

mol solute = ~~(5.0234) / (58.44277)~~
 mol solute = $5.0234 \text{ g} / 58.44277 \text{ g/mol}$
 $\text{mol} = \frac{5.0234 \text{ g}}{58.44277 \text{ g/mol}} = 0.085954 \text{ mol}$

Station #2

foil = 0.3299 g
 Mr. 26.98 g/mol
 Al₂O₃ = 474.46 g/mol

Station #3

water temp = 18.2°C
 atmospheric pressure = 763.0 mmHg
 $P_{\text{total}} = P_{\text{H}_2} + P_{\text{H}_2\text{O}} = 763.0 \text{ mmHg} = P_{\text{H}_2} + 15.673 \text{ mmHg}$
 $P_{\text{H}_2\text{O}} = 15.673 \text{ @ } 18.2^\circ\text{C}$
 $P_{\text{H}_2} = 763.0 \text{ mmHg} - 15.673 \text{ mmHg} = 747.3 \text{ mmHg}$

Station #4

$q = m \times c \times \Delta T$
 30.25 g - NaOH added
 Total solution = 35.0392 g
 Specific heat = 4.184 J/g°C

$\Delta T = 21.1^\circ\text{C} - 4^\circ\text{C} = 17.1^\circ\text{C}$
 $q = 35.0392 \text{ g} \times (4.184 \text{ J/g}^\circ\text{C}) \times (17.1^\circ\text{C})$
 $q = 3095.31 \text{ J} \approx 3090 \text{ J}$

Station #5

33.0 mL
 40.0 mL

Volume of object "C" = 7.0 mL
 $40.0 \text{ mL} - 33.0 \text{ mL} = 7.0 \text{ mL}$

Station #6

Observations: Turned milk-white color - center salt

Reaction Type: Precipitation

Balanced Equation: $\text{Na}_3\text{PO}_4 + \text{BaCl}_2 \rightarrow 2\text{NaCl} +$

Station #7

final buret reading = 19.90 ml

Station #8

pipet & watch glass
 used to transfer a specific amount of liquid - used with 3 different types of methoxy
 used to set & observe experiment on

Station #9

3.88 cm

Station #10

mg = 0.1495 g

MgXOX = 0.2479

Oxygen = 0.0984

$0.0984 / 0.2479 \times 100 = 39.69\%$

mass% = $(0.1495 / 0.2479)(100)$

mass% = $(0.6030)(100)$

mass% of mg = 60.31

Extra Space

CHEM 3A - Lab Practical - VERSION BName: STEVEN MORGAN

Section #: _____

Student ID #: 0184403

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

$$100.1147 - 46.9067$$

$$53.208 = \text{SOLUTION}$$

$$\frac{5.0277}{53.208} \times 100 = 9.4491$$

Station #2

$$\text{EMPTY VILE } 17.2662$$

$$\text{ALUM VILE } 27.1941$$

$$27.1941 - 17.2662 = 9.9279$$

$$\text{MASS OF ALUM } 9.9279$$

$$\text{ACTUAL YIELD}$$

Station #3

$$40.3 \text{ mL}$$

Station #4

- TEMPERATURE WENT DOWN

- EXOTHERMIC

Station #5

$$3.8 \text{ cm}$$

$$38 \text{ mm}$$

Station #6

Observations: NO OBSERVABLE REACTION

Reaction Type: NO REACTION

Balanced Equation: $\text{HCl} + \text{Na}_2\text{CO}_3$ **Station #7**

0.9573 m

10.00 mL VED

1.27 mL $\rightarrow$ 17.12 mL

$$\frac{0.9573}{17.12}$$

0.0559 m

mol sol / L sol

Station #8

- VOLUMETRIC FLASH

- TEST TUBE

Station #9

28.0 mL GRADUATED CYLINDER

21°C FLASH

Station #10

MASS OF OXYGEN 0.1003 g in msxOy

39.69 %

$$\frac{\text{MASS OF msxOy } 0.2527}{0.2527} \times 0.1003$$

Extra Space